

Information Theory for Data Science

Assignment 01

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Politecnico di Torino - Data Science and Engineering

Assignment 1

Introduction to Information Theory and applications to classifiers

1. Alternative Entropy definitions
2. How unpredictable is a sporting event?
3. Can entropy reveal the structure of a text?
4. Maximum entropy under mean constraints
5. Entropy of the function of a variable
6. Statistical Dependence and its impact
7. Which feature is more informative?
8. KL divergence
9. Ternary confusion matrix
10. Build a decision-tree classifier using the Information Gain Ratio.

IMPORTANT

The countdown to the deadlines will start when the final version of Assignment 1, containing all the exercises, is uploaded to Dropbox.

Exercise 1 – Beyond Shannon entropy

Shannon entropy is not the only measure of uncertainty.

Investigate alternative definitions of entropy.

1. Present and explain at least two alternative definitions of entropy, including **Rényi entropy** and **min-entropy**
2. Explain how they differ from Shannon entropy and what aspects of a probability distribution they emphasize.
3. Show how Shannon entropy is related to Rényi entropy and minimum entropy.
4. Consider one probability distribution of your choice and compare numerically the different entropy measures.
5. Discuss real-world examples in which an alternative entropy measure may be more appropriate than Shannon entropy.

Exercise 2 – How unpredictable is a sporting event?

1. Select three sporting events: two with the same number of outcomes and one with a different number.
2. From decimal odds o_i , form $q_i = 1/o_i$ and normalize $p_i = q_i / \sum_j q_j$.
3. Plot the three pmfs and compute $H(X)$.
4. Also compute the normalized entropy $H_{\text{norm}} = H(X) / \log_2 M$.
5. Discuss the results
6. Discuss what changes when the number of possible outcomes M is different.

Exercise 3 – Can entropy reveal text structure? (1/2)

1. Select at least 1000 words of English text, convert to lowercase, remove punctuation, and keep the 26 letters plus space.
2. Estimate the probability distribution of the individual symbols and represent it graphically.
3. Estimate the joint probability distribution of consecutive symbol pairs and represent it using a suitable two-dimensional plot.
4. Randomly shuffle all characters and repeat the previous analysis: estimate and plot the single-symbol and symbol-pair probability distributions.
5. Compare the distributions before and after shuffling and comment on the observed differences.

Exercise 3 – Can entropy reveal text structure? (2/2)

6. For the original text, compute the single-symbol entropy H_1 and the pair entropy H_2 . Report also $H_2/2$ in bits/character.
7. Repeat the entropy calculations for the shuffled text.
8. Verify that shuffling does not significantly change H_1 . What happens instead to $H_2/2$?
9. Explain the results in terms of statistical dependencies between consecutive symbols and discuss what this experiment reveals about the structure of natural language.

Exercise 4 – Maximum entropy under mean constraints

Invent a real-world example representing four possible computational costs (you MUST present an example not only the 4 numerical values). (Order for increasing values of x_i)

1. Compute numerically the MaxEnt pmf for three different values of μ
2. For each case, solve for β , plot the pmf, and compute the entropy.
3. Explain which value of μ produces the uniform distribution.
4. Compare the three distributions and interpret how the mean constraint changes their shape.
5. Discuss if MaxEnt necessarily recovers the true distribution that generated the data

Numerical Solution of the Maximum-Entropy Equation

You must numerically solve the equation generated by the Lagrange optimization to determine β .

For example, in MATLAB you can use:

MATLAB example

```
syms beta  
eqn = ( ... ) == 0,  
beta_sol = vpasolve(eqn,beta,[0 max_range]),
```

Important: max_range must be chosen according to the numerical values used in your example.

Only admissible solutions must be considered:

$$\beta > 0$$

Exercise 5a - Entropy of the function of a variable

Prove that each collision can only reduce entropy

Exercise 5b - Entropy of the Function of a Variable

Provide a **real-world example** of a **non-injective function** (do not use an all-to-one mapping)

$$Y = f(X).$$

Choose realistic values for the probabilities of X , determine the probability distribution of Y , and compare

$$H(X) \quad \text{and} \quad H(Y).$$

Discuss the meaning of the entropy reduction in your example.

Exercise 6 - Statistical Dependence and its impact

Consider two discrete random variables

$$\Omega_X = \{x_1, x_2\}, \quad \Omega_Y = \{y_1, y_2\}.$$

Construct two **real-world examples**:

- **Example A:** X and Y are statistically independent.
- **Example B:** X and Y are statistically dependent.

For each example:

1. Clearly explain the real-world meaning of X and Y .
2. Define a realistic joint pmf $p(x, y)$ and represent it as a 2×2 table.
3. Compute the marginal pmfs $p(x)$ and $p(y)$.
4. Compute

$$H(X), \quad H(Y), \quad H(X, Y).$$

Exercise 6 - Statistical Dependence and its impact

For both real-world examples introduced in the previous slide:

1. Compute the conditional entropies

$$H(X|Y), \quad H(Y|X).$$

2. Verify the chain rules

$$H(X, Y) = H(Y) + H(X|Y)$$

and

$$H(X, Y) = H(X) + H(Y|X).$$

3. Compute the mutual information

$$I(X, Y) = H(X) + H(Y) - H(X, Y).$$

4. Compare the two examples and explain the results in terms of the information that observing one variable provides about the other.

In particular, verify that

$X \text{ and } Y \text{ independent} \implies I(X, Y) = 0$

and compare this result with the statistically dependent case.

Exercise 7 – Which feature is more informative?

Invent a **real-world** dataset for a **binary classification problem** containing

- one binary target variable X ,
- at least three discrete input features Y_1, Y_2, Y_3 ,
- at least 10 observations.

For each feature Y_i :

1. Estimate the joint pmf $p(x, y_i)$ and the corresponding marginal pmfs.
2. Compute

$$H(X), \quad H(X|Y_i).$$

3. Compute the mutual information

$$I(X, Y_i) = H(X) - H(X|Y_i).$$

4. Verify numerically that

$$I(X, Y_i) = H(X) + H(Y_i) - H(X, Y_i).$$

Exercise 7 – Interpretation

For the dataset introduced in the previous slide:

1. Compute the normalized information gain

$$\frac{I(X, Y_i)}{H(X)}$$

for each feature.

2. Identify the most informative and the least informative feature about the target X .

Exercise 8 – Comparing probability models with KL divergence

Consider a discrete random variable with at least five possible outcomes.

Invent a **real-world example** and define an empirical pmf P representing observed data.

Then define three different candidate models

$$Q_1, \quad Q_2, \quad Q_3.$$

1. Plot the four pmfs P, Q_1, Q_2, Q_3 .
2. For each candidate model compute

$$D_{\text{KL}}(P||Q_i) = \sum_x p(x) \log_2 \frac{p(x)}{q_i(x)}.$$

3. Identify the model having the smallest KL divergence from the empirical distribution.
4. Verify numerically that

$$D_{\text{KL}}(P||Q_i) \geq 0.$$

Exercise 8 – Interpretation of KL divergence

Using the probability models introduced in the previous slide:

1. Compute also the reverse divergences

$$D_{\text{KL}}(Q_i \| P).$$

2. Verify if

$$D_{\text{KL}}(P \| Q_i) = D_{\text{KL}}(Q_i \| P).$$

and discuss if the KL divergence is a distance

3. Discuss what a small value of

$$D_{\text{KL}}(P \| Q_i)$$

tells us about the candidate model Q_i .

4. Discuss what happens if

$$p(x) > 0 \quad \text{but} \quad q_i(x) = 0$$

for one possible outcome x .

Exercise 9 – Ternary confusion matrix

Consider a classification problem with three possible classes:

$$\Omega_C = \{c_1, c_2, c_3\}.$$

1. Construct the corresponding confusion matrix.
2. Derive an expression for the overall classification error probability in terms of the confusion-matrix entries.

Exercise 10: Build a decision-tree classifier using the Information Gain Ratio.

Consider the following training set with three numerical features

$$\underline{x} = (x_1, x_2, x_3)$$

and binary class $c \in \{0, 1\}$.

#	x_1	x_2	x_3	c
1	20	60	80	0
2	80	20	60	1
3	40	80	20	1
4	80	80	40	0
5	20	40	80	1
6	60	60	40	1
7	20	40	20	0
8	20	60	20	0

#	x_1	x_2	x_3	c
9	80	60	60	0
10	80	60	20	0
11	60	20	40	0
12	80	80	20	0
13	40	40	60	1
14	20	80	60	0
15	80	80	60	0

Goal: build a decision-tree classifier using the Information Gain Ratio.

For each level of the decision tree:

- list the candidate thresholds considered for each available feature,
- for each threshold, show all calculations required to obtain the Information Gain Ratio,
- for each feature, select the threshold with the largest Information Gain Ratio,
- compare the best thresholds of all available features and select the feature and threshold used for the split.

Repeat the same procedure at every subsequent level of the tree.

- draw the complete decision-tree classifier,
- apply the classifier to all vectors in the training set,
- construct the corresponding confusion matrix,
- compute the false alarm probability and the missed detection probability,
- comment the result

Apply the classifier to the following 10 test vectors:

#	x_1	x_2	x_3	c
1	22	48	25	0
2	33	70	40	0
3	28	32	72	1
4	65	35	42	0
5	55	42	62	1
6	45	65	25	1
7	68	75	78	1
8	75	55	35	0
9	25	72	70	0
10	60	68	30	1

- compare the predicted class with the true class for each test vector,
- construct the corresponding confusion matrix,
- compute the false alarm probability and the missed detection probability,
- comment the result
- if some test vectors are misclassified, discuss possible reasons based on the training set and on the decision boundaries learned by the tree.

Assignment delivery: to be confirmed

Final version assigned on 07/10/2026

Delivery by

- 28/10/2026, 11.59 PM: +2 point
- 07/11/2026, 11.59 PM: 0 points
- Later: not accepted